# Position Control of Permanent Magnet Synchronous Motors

## Introduction

Position control of Permanent Magnet Synchronous Motors (PMSMs) is a critical function in modern motion control systems, facilitating precise control of position, speed and torque. PMSM servo drives are widely used for position control applications in robotics, CNC machines and industrial automation due to their high efficiency, torque density and superior dynamic response.

PMSM position control solutions employ Field-Oriented Control (FOC), an advanced vector control technique that decouples motor currents into torque and flux components, enabling efficient and dynamic motor control. A high-resolution position sensor provides real-time rotor position feedback, facilitating accurate control.

Modern motion control solutions leverage advanced algorithms, real-time feedback control and digital signal processing capabilities to achieve precise and dynamic position control. The dsPIC33A family of Digital Signal Controllers (DSCs), with its signal processing capabilities, advanced peripherals and low-latency responsiveness, makes it ideal for real-time applications requiring fast and accurate control.

## Features

Key features of the Position Control solution include:

- Field-Oriented Control (FOC): Efficient torque and speed control
- High-resolution Position Feedback Interface (Absolute or Incremental) for Real-Time Rotor Position Feedback
- Motion Profile Generation: Supports S-Curve (Third order) and Trapezoidal (Second order) motion profiles
- High-bandwidth Control Loop: Ensures accurate, fast and stable response to dynamic load variations
- Autotuning for Current and Speed PI Control Loops
- Homing Operation: Initializes position to a known reference point at startup
- Integrated Overcurrent and Overvoltage Protection
- Communication Protocol: MODBUS

## Applications

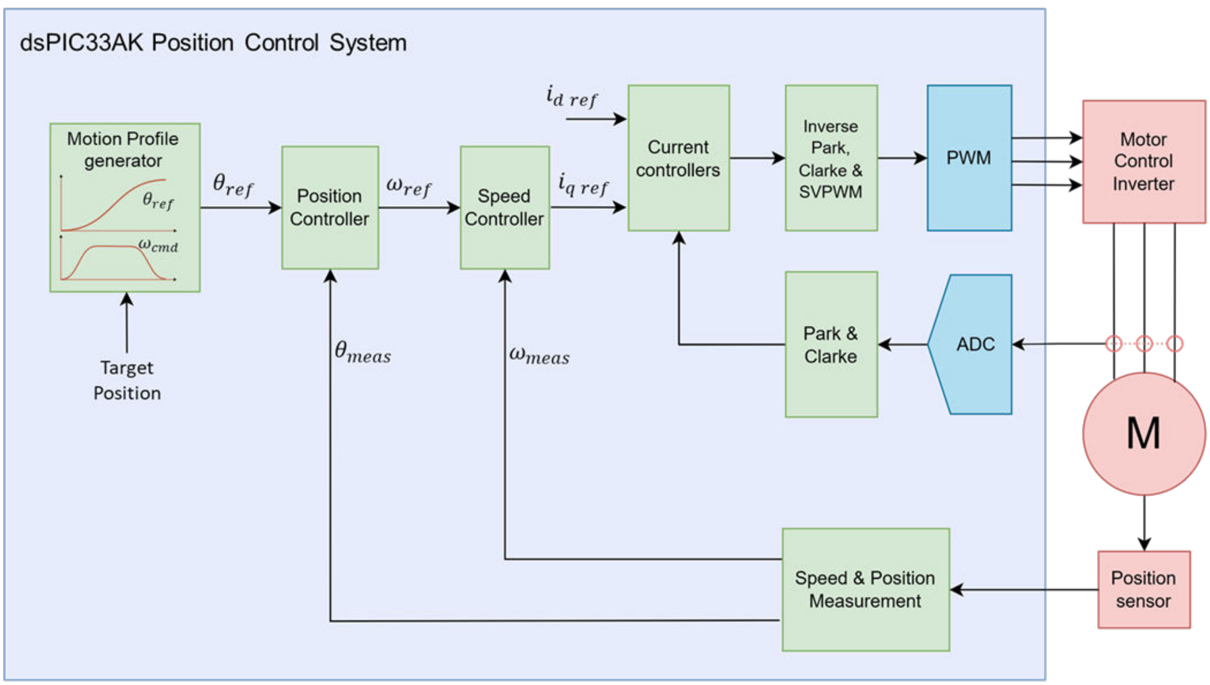
Position Control Solutions has the following applications:

- Industrial automation: Conveyor systems, pick-and-place machines
- Robotics: Precise joint positioning and collaborative robot applications
- CNC machines: High-accuracy spindle and axis control
- Medical devices: Motorized prosthetics and surgical robotics

## Position Control of Permanent Magnet Synchronous Motors

Position control primarily consists of position and speed feedback, three cascaded control loops—current, speed, and position—and a motion profile generator. Figure 1 illustrates the block diagram of the position control system.

**Figure 1.** Control Block Diagram




### Cascaded Control Loops

The position control comprises three main cascaded control loops:

1. Current Loop (FOC)
   - The innermost loop is implemented using Field-Oriented Control (FOC) to regulate the motor currents.
   - This control structure uses two PI control loops to independently regulate the d-axis and q-axis currents. The q-axis PI controller controls the torque-producing current, while the d-axis PI controller manages the flux-producing current.
   - The speed controller generates the reference (desired) q-axis current.
   - The PI controller outputs are converted into voltage commands, which are then processed by the Space Vector Pulse Width Modulation (SVPWM) algorithm to generate optimized switching signals for the inverter, ensuring smooth and efficient motor operation.

2. Speed Control Loop
   - This loop regulates motor speed to ensure smooth and stable speed tracking.
   - The reference (desired) speed is generated from the position loop and is processed by a speed PI controller, which outputs the required torque demand.
3. Position Loop (Outer Loop)
   - The outermost loop ensures the rotor reaches and maintains the desired position accurately.
   - A Proportional-Integral-Derivative (PID) controller regulates position error.
   - The reference position is compared with the actual position obtained from a position sensor and the computed error determines the desired speed.
   - Filtered derivative term is implemented to reduce sensitivity to noise. The PID controller transfer function is given by:

$$\frac{\omega_{ref}(s)}{E(s)} = k_p + k_i \frac{1}{s} + k_d \frac{s}{1 + \tau_{d,f} s}$$

   where:
   - $k_p$ = Proportional gain
   - $k_i$ = Integral gain
   - $k_d$ = Derivative gain
   - $\tau_{d,f}$ = Derivative filter time constant

**Position and Speed Feedback**

Position and speed feedback systems offer high-resolution position data and perform calculations for speed estimation. The available options are as follows:

- Absolute encoders and magnetic position sensors: These devices provide direct feedback on the rotor's position.
- Incremental encoders with Hall sensors: Incremental encoder signals (Q_A, Q_B, Q_INDX) are combined with Hall sensor outputs (HA, HB, HC) to enable absolute position referencing.

The Quadrature Encoder Interface (QEI) module in the dsPIC DSC can directly acquire mechanical position data from incremental encoders, supporting four inputs: Phase A, Phase B, Index, and Home. Absolute encoders using T-Format and A-Format protocols can be managed through the SPI or UART modules. Additionally, the dsPIC33A DSC is compatible with the Bidirectional Serial Synchronous (BiSS) Encoder Interface module, which supports BiSS-C, SSI, and EnDAT-based protocols.

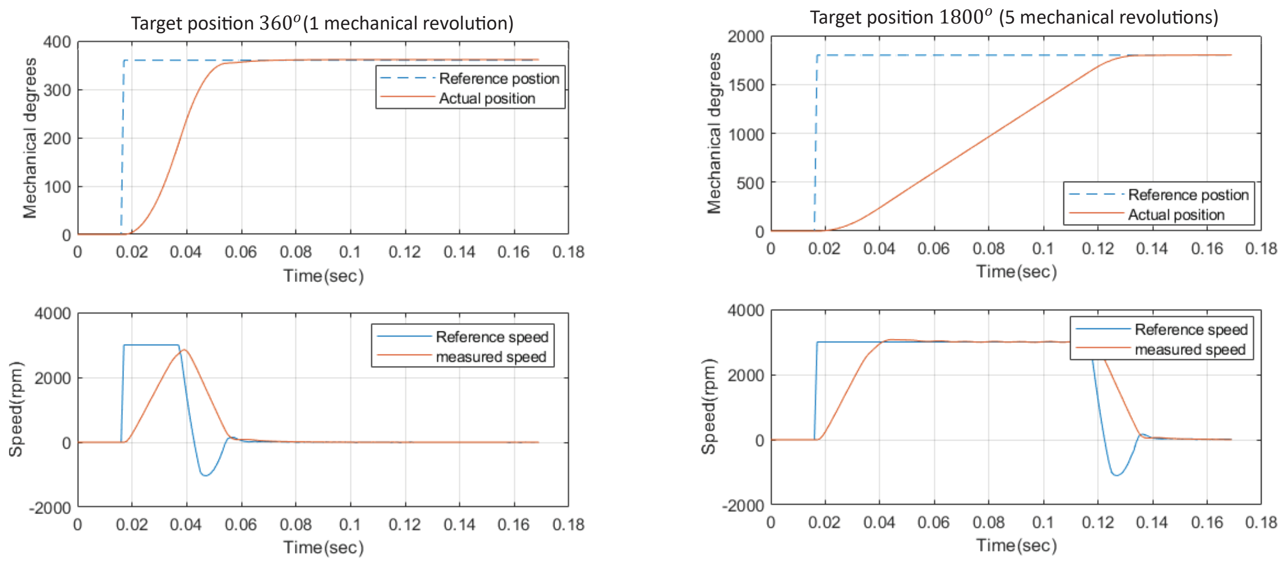
**Motion Profile Generation**

Motion profile specifies the desired velocity, acceleration and position of the system over time, ensuring smooth operations and minimizing the electrical and mechanical stress.

The solution includes the following common motion profiles:

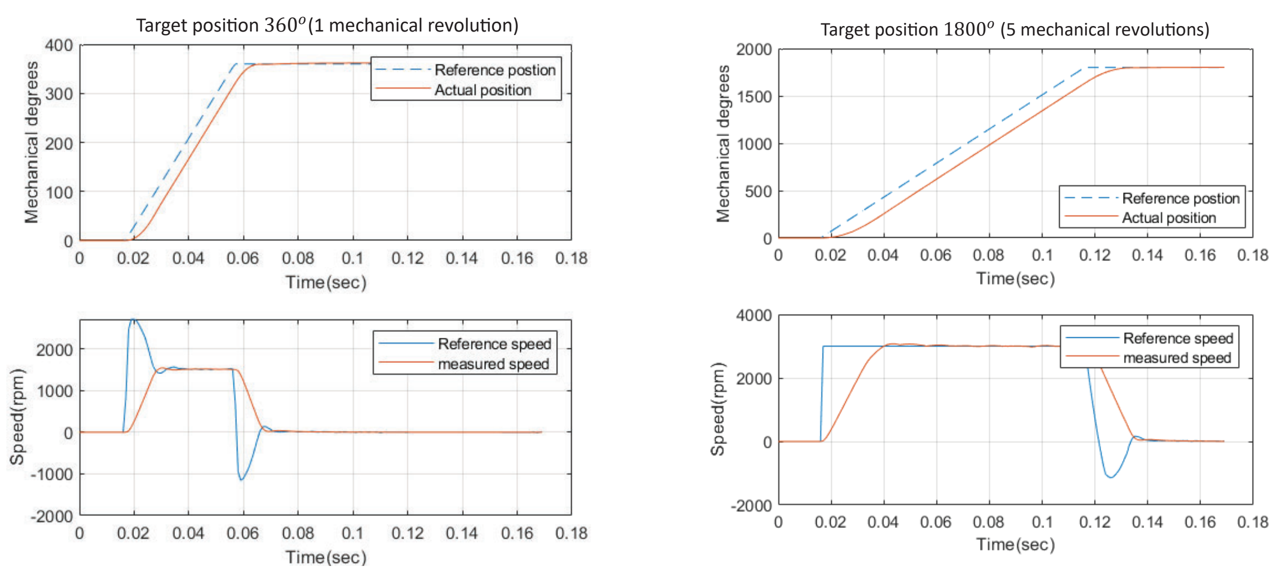
1. Flat profile (Step response):
   - The system instantaneously moves to the target position without velocity or acceleration constraints, as shown in Figure 2.
   - This causes high mechanical stress and is generally unsuitable for most of the motion control applications.

**Figure 2.** Flat Motion Profile



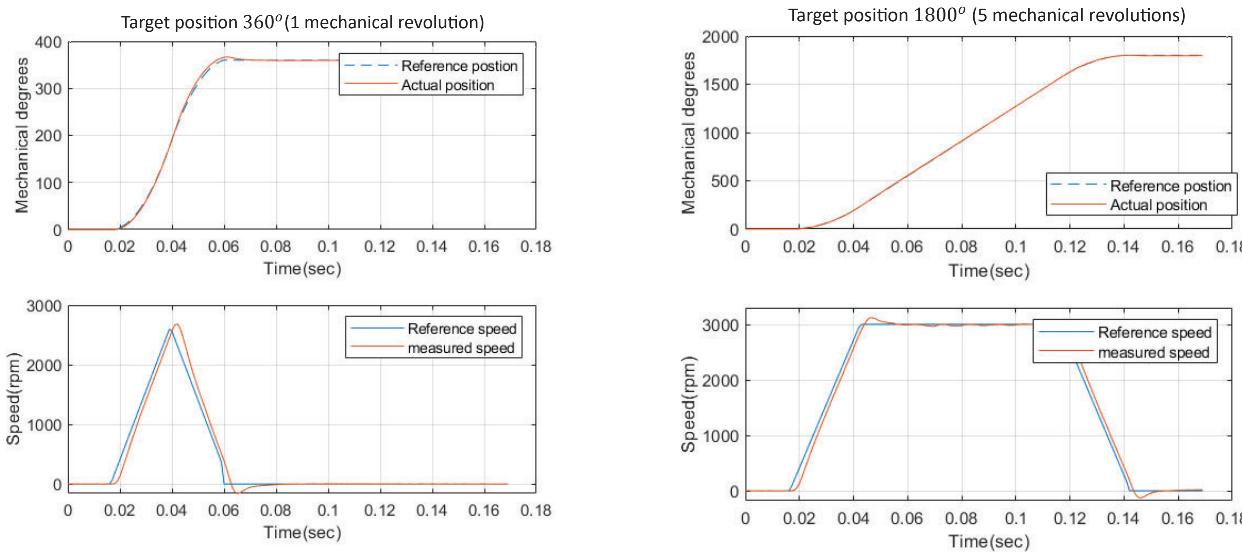

2. Ramp (First order) profile:
   - The system moves at a constant velocity and acceleration peaks at the start and end of the motion, as shown in Figure 3.
   - It is simple to implement but abrupt acceleration causes mechanical stress.

**Figure 3.** Ramp Motion Profile



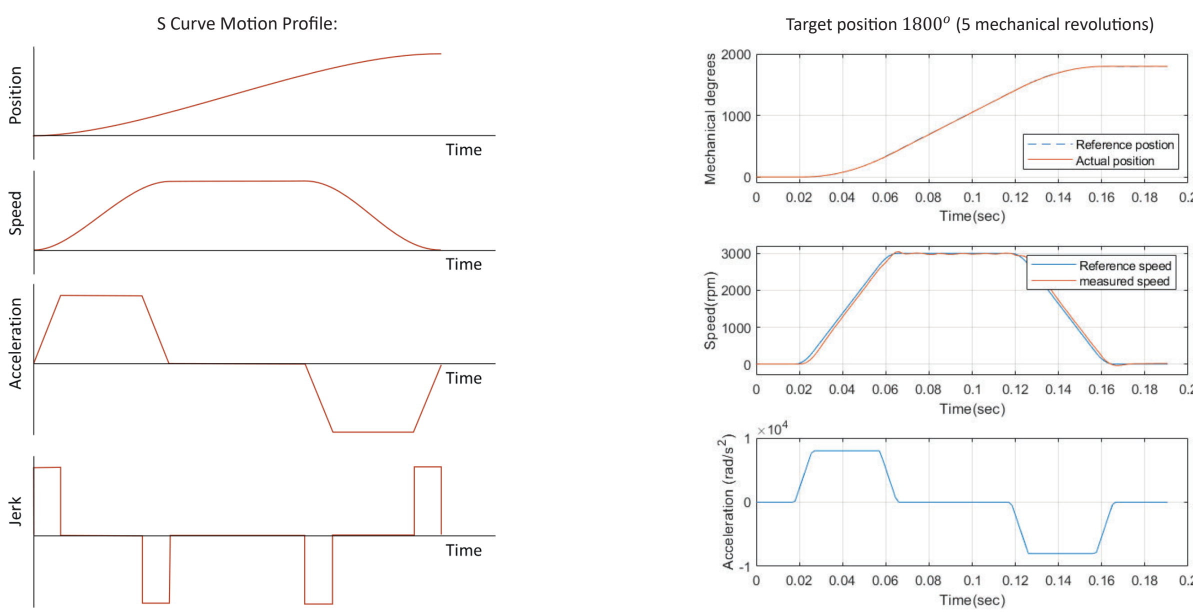

3. Trapezoidal (Second order) profile:
   - The profile consists of three phases: acceleration, constant velocity and deceleration, as shown in Figure 4.
   - This requires target position, maximum velocity, maximum acceleration and maximum deceleration information.
   - Smooth transitions compared to ramp profile but still result in sudden changes in acceleration.

**Figure 4.** Second Order Motion Profile




4. S-Curve (Third order) profile:
   - This profile introduces a gradual change in acceleration, effectively reducing jerk (rate of change of acceleration).
   - It consists of the following distinct phases: acceleration ramp-up, constant acceleration, acceleration ramp-down, constant velocity and symmetric deceleration phases, as shown in Figure 5.
   - The smooth transitions between these phases help minimize mechanical stress, enhance motion smoothness and improve system longevity. This profile is well-suited for precision applications where gentle motion is critical.
   - Required inputs include: target position, maximum velocity, maximum acceleration, maximum deceleration and maximum jerk.

**Figure 5.** Third Order Motion Profile



The results presented in this document are based on the Leadshine ELVM6020V24FH-B25-HD (24 V, 200 W) servo motor, equipped with a 2500 PPR incremental encoder. The following table summarizes its electrical and mechanical specifications.

**Table 1.** Motor Specifications

| Motor Specifications | |
|---|---|
| Rated voltage | 24V |
| Rated speed | 3000 rpm |
| Rated current | 10A |
| Back EMF constant | 4.1 mV/rpm |
| Inertia | $0.29 \times 10^{-4}$ kg m$^2$ |
| Resistance | 0.18Ω |
| Inductance | 0.33 mH |
| Motor poles | 10 |
| Encoder resolution | 2500 ppr |

## Position Control Demonstration Setup

The following figures show the demonstration setup for Position Control of a PMSM servo motor. The MCLV-48V-300W Development Board (EV18H47A) along with dsPIC33AK128MC106 Motor Control Dual In-line Module (EV68M17A) is used for the development.

**Figure 6.** Demonstration Setup Showing Angular Position Control Using a Mechanical Lever

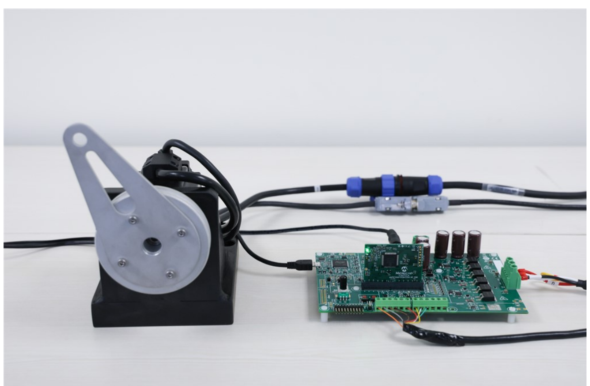
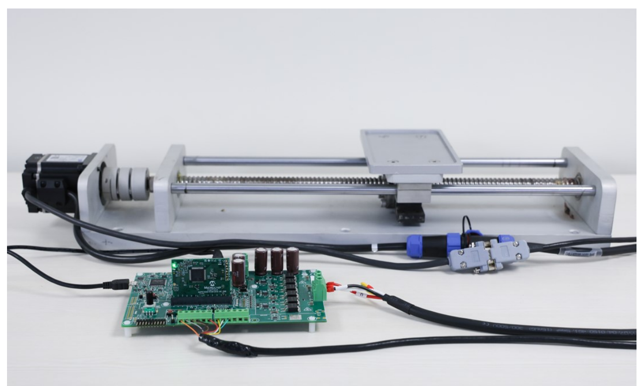


**Figure 7.** Setup Demonstrating Linear Motion Control by Driving a Lead Screw Mechanism



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